

International Islamic University Chittagong
Center for General Education
Midterm Exam- Spring 2026
Course Code: URED- 3503 (URED- 2101 for LLB)
Course Title: Political Thoughts and Social Behavior

Time: 1.30 Hours

Full Marks: 30

(All questions are of equal value)

SL	Questions	Marks	CLO	Bloom's taxonomy domain
1	Define Islamic politics and explain how its basic principles shape the Islamic political system.	10	2	Remember& Create
2	What is meant by <i>Shariah</i> ? Explain some key features of a constitution that follows <i>Shariah</i> .	10	2	Remember& Create
3	Analyze the qualities and main duties of chief executive of an Islamic State. Or How does the concept of accountability before Allah and the people strengthen an Islamic state?	10	2	Analyze

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)

Program: B.Sc. in CSE: Semester: 5th

Mid-Term Examination, Spring-2026

Course Code: CSE 3523

Course Title: Microprocessor, Microcontroller and Embedded System

Time: 1 hour 30 minutes

Full Marks: 30

Answer the following Three (3) questions. Each question carries 10 marks.

		Questions	CLOs	Marks																					
1	a)	Why is the Data Bus Bi-directional while the Address Bus is Unidirectional? Calculate the minimum number of address lines required to access 16 MB of memory?	CO1	2																					
1	b)	What are the four fields of a basic assembly language statement? Write an instruction that uses all four fields and label each part clearly	CO2	2																					
1	c)	Draw the internal architecture of 8086 and explain the role of: segment Registers ,Instruction Pointer ,Instruction Queue ,ALU and Flag Register	CO2	6																					
2	a)	Explain the purpose and function of the directives .model, .stack, .data, .code, PROC, ENDP, and end in an Assembly language program	CO2	5																					
		OR																							
2	a)	Draw the format of the 16-bit Flag Register of 8086 and explain the role of status flags and control flags, Find the value of the status after the arithmetic operation. MOV AL, 15H ADD AL, 2AH	CO2	5																					
2	b)	I. MOV AX, [BX][SI] II. MOV AX,1H[BX][SI] III. MOV CS,[2004] IV. MOV 82H, AL V. MOV [AX], [2001] Identify the addressing mode for each instruction Above. If invalid, explain why. Correct the instruction and update the register and memory value. <table><tr><th>Initial Value</th></tr><tr><td>AX=2002</td></tr><tr><td>BX=2003</td></tr><tr><td>CX=2456</td></tr><tr><td>DX=0002</td></tr><tr><td>SI=0001</td></tr><tr><td>DI=0004</td></tr></table> <table><tr><th>Offset address</th><th>Data</th></tr><tr><td>2001H</td><td>11H</td></tr><tr><td>2002H</td><td>02H</td></tr><tr><td>2003H</td><td>99H</td></tr><tr><td>2004H</td><td>32H</td></tr><tr><td>2005H</td><td>33H</td></tr><tr><td>2006H</td><td>44H</td></tr></table>	Initial Value	AX=2002	BX=2003	CX=2456	DX=0002	SI=0001	DI=0004	Offset address	Data	2001H	11H	2002H	02H	2003H	99H	2004H	32H	2005H	33H	2006H	44H	CO2	5
Initial Value																									
AX=2002																									
BX=2003																									
CX=2456																									
DX=0002																									
SI=0001																									
DI=0004																									
Offset address	Data																								
2001H	11H																								
2002H	02H																								
2003H	99H																								
2004H	32H																								
2005H	33H																								
2006H	44H																								
3	a)	Explain with suitable examples and compare RISC, CISC, and EPIC architectures in terms of performance, complexity, and efficiency.	CO1	5																					
3	b)	MOV AX,FF0EH XOR AX,FC00H SHL AL,4 What will be the final value of register and carry? Show the all shifting stages.	CO2	5																					
		OR																							
3	b)	Calculate the displacement for a jump from address 0007H to target address 0200H. Is this a short jump or a near jump? Justify. Write an assembly program to compute the sum: 5+10+15+ +90	CO2	5																					

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE, Midterm Examination, Spring 2026

Course Code: CSE-3529

Course Title: **System Analysis and Design**

Total marks: 30, Time: 1 hour 30 minutes

[Answer all the questions. Figures in the right-hand margin indicate full marks]

1. a) Different organizational levels require different types of information. Analyze how different types of information support decision-making at each management level. Provide examples for each type. 5 CO1

OR (of 1a).
Compare Decision-Support Systems (DSS) and Management Information Systems (MIS). How do their roles, data usage, and types of decisions they support differ within an organization?
- b) Discuss the major roles and responsibilities of a systems analyst. How do the essential qualities of a good systems analyst help them perform these duties effectively? 5 CO2
2. a) Explain the Systems Development Life Cycle (SDLC) and describe its main phases. How does each phase contribute to the successful development and implementation of an information system? Provide examples for each phase. 5 CO2
- b) Briefly describe Waterfall model and Spiral model. Which model is better? Why? 5 CO2
3. a) A company wants to expand globally and uses SWOT analysis? How does SWOT analysis support IT system planning? How does strategic planning influence system requests? 5 CO2
- b) Describe the steps involved in performing an economic feasibility analysis of a project. 5 CO2

OR (of 3b)

Find whether the following project is economically feasible or not. A government plans a national identity system with an initial cost of \$500,000. Expected annual savings (benefits) are:

- Year 1: \$150,000
- Year 2: \$180,000
- Year 3: \$200,000
- Year 4: \$220,000

rate of return= 10%

Bismillahir Rahmanir Rahim
 International Islamic University Chittagong
 Department of Computer Science and Engineering
Mid-Term Examination, Spring-2026
 Course: **CSE-3521 (Computer Architecture)**
 Time: **1 hour and 30 Minutes** Marks: **30**

[Answer all *three* questions; Figures in the right-hand margin indicate Marks]

- Q1. a) A server runs two processes simultaneously. Process A reports 48.6u 22.4s 3:30 and Process B reports 30.2u 8.8s 2:15. Which process utilized the CPU more efficiently? Justify with calculation. 5 CO2
- b) A 32-bit processor runs at 5 GHz with two compilers generating instruction mixes across classes A (CPI=1), B (CPI=2), and C (CPI=3). Calculate the execution time for each compiler and determine which one performs faster, justifying why raw instruction count alone is a misleading performance metric. 5 CO2

- Q2. a) A program adds \$t1 and \$t2 storing the result in \$t3, then stores \$t3 in memory at offset 100 from \$s0, and finally jumps to address 0x000A0000. Convert all three instructions into 32-bit binary machine code by showing their assembly codes. 5 CO2
- Register encodings:** \$t1=1001, \$t2=1010, \$t3=1011, \$s0=10000
 Opcodes/funct: add funct=100000, storing opcode=101011, Jump opcode=000010, all R-type opcode=000000, shamt=00000
- b) If the Memory Access stage were removed from the instruction execution pipeline, which MIPS instructions would break and why? 5 CO1

Or

- Q2. a) Decode the following assembly into 32-bit binary machine code, and explain why j 0x000A0000 cannot store its full target address in a single instruction word: 5 CO2

sub \$t3, \$s1, \$s2
 sw \$t3, 100(\$s0)
 j 0x000A0000

- b) A branch instruction and a jump instruction both alter the program flow. Compare the addressing modes used by each, and explain why MIPS uses different addressing modes for these two instructions. 5 CO1

- Q3. a) Two engineers debate which number should be the Multiplicand and which should be the Multiplier when computing 6×5 . Engineer A places 6 as Multiplicand and 5 as Multiplier, while Engineer B does the opposite. Using the refined (final) version of the multiplication algorithm, trace both, draw the flowchart diagram for one iteration, and determine whether the choice affects the number of additions performed. 5 CO3
- b) Represent -0.625_{10} in IEEE 754 Single Precision format. Identify what changes in the exponent and sign bit if the number were positive, and explain why the mantissa remains unaffected by the sign. 5 CO2

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science & Engineering (CSE)

BSc in CSE, Mid-Term Examination, Spring 2026

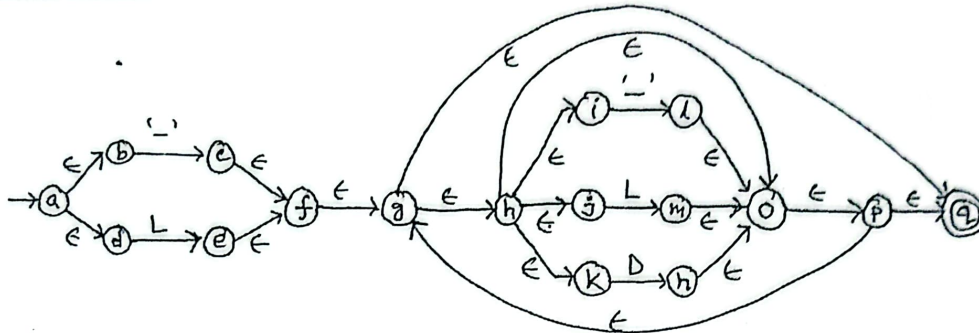
Course Code: CSE-3527 Course Title: Compiler

Full Marks: 30 Time: 1 Hour 30 minutes

[Answer all the following questions. Some questions may have an option.
(Figures at right margin illustrate the marks (M), course objectives (COs), and bloom's taxonomy (DL))]

No.	Questions	M	COs	DL
1. a)	Define the compiler using an appropriate block diagram and list some useful compiler-construction tools.	1	CO1	U
	Or			
	Explain the analysis of the source program with different phases	1	CO1	U
b)	Consider the following C program: <pre>int main() { int a = 10, b = 5; float c; c = a + b * 2; printf("%f\n", c); return 0; }</pre>	9 (4+1+2+2)	CO1	Ap
	Analyze and step-by-step translate the above C program through the different compiler phases. In your analysis and translation:			
	i) Draw a labeled diagram of compiler phases, and annotate the output produced at each stage for the given program			
	ii) Construct the symbol table, showing all identifiers along with their relevant pattern name			
	iii) Define the syntax for the given expressions in the above program using a suitable context-free grammar (CFG)			
	iv) Perform lexical analysis by identifying each lexeme and classifying it into appropriate tokens, and generate a table to illustrate the lexemes, patterns, and tokens			
2.a)	What is the basic difference between syntax definition and syntax-directed translations in compiler construction? Give the syntax definition (CFG) to define the numerical constant (Including integers, fractions and exponential numbers) of the C programming language and justify how the numerical constant string "35.27E-5" can be generated by your syntax definition grammar using the left-most derivation or parse tree.	4 (1+2+1)	CO2	Ap
	Or,			
	Elucidate the ambiguity of CFG. Construct the unambiguous CFG for arithmetic expressions of integers and identifiers with the four binary operators {+, -, *, /} for the C programming language. Justify that your grammar is correct for the input string "a*2+3-5" using the left-most derivation and parse tree.	4 (1+3)	CO2	Ap
b)	Explain all types of grammar to define a language with rules and restrictions by emphasizing the production ($\alpha \rightarrow \beta$) to differentiate their level [where α is production head and β is production body]. Give a simple example of each type of grammar. Define the inaccessible token and useless non-terminal using a simple example.	4 (3+1)	CO2	Ap

- c) Explain the main task of the lexical analyzer for the tokenization of the source program with the appropriate block diagram along with lexical errors and possible error-recovery actions. 2
- 3.a) Write the regular expression and draw the nondeterministic finite automata (NFA), deterministic finite automata (DFA) transition diagram, and ϵ -NFA to define the numerical constants in the C programming language. 5 (3x1+2) CO2 Cr
- b) Consider the following ϵ -NFA to define an identifier of the C programming language ($\Sigma = \{ _ , L, D \}$) [Where L represents all English letters ('a-zA-Z') and D represents all digits ('0-9')]. Convert it into the DFA using the ϵ -closure technique step-by-step with the possible DFA state minimization: 5 CO2 Ap



Or,

Write the 5-tuples of finite automata and contrast DFA and NFA. Describe the language ($\Sigma = \{x, y\}$), NFA and ϵ -NFA denoted by the regular expression: $x(x|y)^*xxy\#$

5 (2+3) CO2 Ap

END

International Islamic University Chittagong
Department of Computer Science and Engineering (CSE)

B.Sc. Engineering in CSE
 Mid-term Examination, Spring 2026

Course Code: EEE-2421

Course Title: Electrical Drives and Instrumentation

Time: 1.5 hours

Full Marks: 30

Answer all the questions. The figures in the right-hand margin indicate full marks
 Course Outcome (COs) and Bloom's Level (BL) are mentioned in additional Columns

1 (a)	What is Eddy current loss? Derive the expression for the induced voltage from a time-varying magnetic field, explaining with proper diagrams.	CO1	An	5
1 (b)	i. Explain the Intermittent periodic duty class with starting and braking. ii. A coil that has 700 turns develops an average induced voltage of 50 V. What must be the change in the magnetic flux that occur to produce such a voltage if the time interval for this change is 0.7 seconds?	CO3	Ap	2+3
OR				
1(a)	Analyze the linear DC Machine with suitable equations and figures. Also explain its starting problems.	CO1	An	5
1(b)	i. Explain the Intermittent periodic duty class with starting. ii. The magnetic flux linked with a coil having 300 turns is changed from 2.4 Wb to 5 Wb in 2.2 seconds. Calculate the induced e.m.f in the coil.	CO3	Ap	2+3
2 (a)	i. State the condition for maximum efficiency in a transformer. ii. Describe the iron losses of a transformer.	CO1	R	3+2
2 (b)	A 50 kVA, 4400/220 V transformer has $R_1 = 3.45 \Omega$, $R_2 = 0.009 \Omega$. The values of reactance are $X_1 = 5.2 \Omega$ and $X_2 = 0.015 \Omega$. Calculate for the transformer: (i) equivalent resistance as referred to both primary and secondary (ii) equivalent reactance as referred to both primary and secondary (iii) equivalent impedance as referred to both primary and secondary.	CO3	An	5
3 (a)	Describe the operation of DC Motor with suitable figures.	CO1	U	5

3 (b)	i. A shunt generator delivers 450A at 230V and the resistance of the shunt field and armature are 50Ω and 0.03Ω respectively. Now, calculate the generated e.m.f ii. A 250-V shunt motor runs at 1000 r.p.m. at no-load and takes 8 A. The total armature and shunt field resistances are respectively 0.2Ω and 250Ω. Calculate the speed when loaded and taking 50 A. Assume the flux to be constant.	CO3	Ap	5
-------	--	-----	----	---